

EXTENDS *Naturals*

This model isolates the race between an idle decision from one tab and a heartbeat carrying newer human activity from a sibling tab. One activity advance is enough to exercise both serializations without obscuring the graph.

VARIABLES

live,
everRevoked,
lastActive,
idleExpected,
idleObserved,
idleHandled,
heartbeatPending,
inputUsed,
activityAdvancedAfterObservation,
lastOutcome

vars $\triangleq$ $\langle$ *live*, *everRevoked*, *lastActive*, *idleExpected*, *idleObserved*,
idleHandled, *heartbeatPending*, *inputUsed*,
activityAdvancedAfterObservation, *lastOutcome* $\rangle$

MaxActivity $\triangleq$ 1

Init $\triangleq$

$\wedge$ *live* = TRUE
 $\wedge$ *everRevoked* = FALSE
 $\wedge$ *lastActive* = 0
 $\wedge$ *idleExpected* = 0
 $\wedge$ *idleObserved* = FALSE
 $\wedge$ *idleHandled* = FALSE
 $\wedge$ *heartbeatPending* = FALSE
 $\wedge$ *inputUsed* = FALSE
 $\wedge$ *activityAdvancedAfterObservation* = FALSE
 $\wedge$ *lastOutcome* = "ready"

ObserveIdle $\triangleq$

$\wedge$ *live*
 $\wedge$ $\neg$ *idleObserved*
 $\wedge$ *idleObserved'* = TRUE
 $\wedge$ *idleExpected'* = *lastActive*
 $\wedge$ UNCHANGED $\langle$ *live*, *everRevoked*, *lastActive*, *idleHandled*,
heartbeatPending, *inputUsed*,
activityAdvancedAfterObservation, *lastOutcome* $\rangle$

$$\begin{aligned}
\text{HumanInput} &\triangleq \\
&\wedge \text{live} \\
&\wedge \neg \text{inputUsed} \\
&\wedge \text{inputUsed}' = \text{TRUE} \\
&\wedge \text{heartbeatPending}' = \text{TRUE} \\
&\wedge \text{UNCHANGED } \langle \text{live}, \text{everRevoked}, \text{lastActive}, \text{idleExpected}, \\
&\quad \text{idleObserved}, \text{idleHandled}, \\
&\quad \text{activityAdvancedAfterObservation}, \text{lastOutcome} \rangle
\end{aligned}$$

$$\begin{aligned}
\text{AcceptHeartbeat} &\triangleq \\
&\wedge \text{live} \\
&\wedge \text{heartbeatPending} \\
&\wedge \text{lastActive} < \text{MaxActivity} \\
&\wedge \text{lastActive}' = \text{lastActive} + 1 \\
&\wedge \text{heartbeatPending}' = \text{FALSE} \\
&\wedge \text{activityAdvancedAfterObservation}' = \\
&\quad (\text{activityAdvancedAfterObservation} \vee \\
&\quad (\text{idleObserved} \wedge \text{lastActive} + 1 > \text{idleExpected})) \\
&\wedge \text{lastOutcome}' = \text{"heartbeat-accepted"} \\
&\wedge \text{UNCHANGED } \langle \text{live}, \text{everRevoked}, \text{idleExpected}, \text{idleObserved}, \\
&\quad \text{idleHandled}, \text{inputUsed} \rangle
\end{aligned}$$

$$\begin{aligned}
\text{RejectHeartbeat} &\triangleq \\
&\wedge \neg \text{live} \\
&\wedge \text{heartbeatPending} \\
&\wedge \text{heartbeatPending}' = \text{FALSE} \\
&\wedge \text{lastOutcome}' = \text{"heartbeat-rejected"} \\
&\wedge \text{UNCHANGED } \langle \text{live}, \text{everRevoked}, \text{lastActive}, \text{idleExpected}, \\
&\quad \text{idleObserved}, \text{idleHandled}, \text{inputUsed}, \\
&\quad \text{activityAdvancedAfterObservation} \rangle
\end{aligned}$$

$$\begin{aligned}
\text{RunIdleRevocation} &\triangleq \\
&\wedge \text{idleObserved} \\
&\wedge \neg \text{idleHandled} \\
&\wedge \text{idleHandled}' = \text{TRUE} \\
&\wedge \text{IF } \text{live} \quad \wedge \text{lastActive} = \text{idleExpected} \\
&\quad \text{THEN } \wedge \text{live}' = \text{FALSE} \\
&\quad \quad \wedge \text{everRevoked}' = \text{TRUE} \\
&\quad \quad \wedge \text{lastOutcome}' = \text{"revoked"} \\
&\quad \text{ELSE } \wedge \text{UNCHANGED } \langle \text{live}, \text{everRevoked} \rangle \\
&\quad \quad \wedge \text{lastOutcome}' = \text{"fenced"} \\
&\wedge \text{UNCHANGED } \langle \text{lastActive}, \text{idleExpected}, \text{idleObserved}, \\
&\quad \text{heartbeatPending}, \text{inputUsed}, \\
&\quad \text{activityAdvancedAfterObservation} \rangle
\end{aligned}$$

$$\text{Next} \triangleq$$

$\vee \text{ObserveIdle}$
 $\vee \text{HumanInput}$
 $\vee \text{AcceptHeartbeat}$
 $\vee \text{RejectHeartbeat}$
 $\vee \text{RunIdleRevocation}$

$$\text{Spec} \triangleq \text{Init} \wedge \Box[\text{Next}]_{\text{vars}}$$

$$\begin{aligned}
\text{TypeOK} \triangleq & \\
& \wedge \text{live} \in \text{BOOLEAN} \\
& \wedge \text{everRevoked} \in \text{BOOLEAN} \\
& \wedge \text{lastActive} \in 0 \dots \text{MaxActivity} \\
& \wedge \text{idleExpected} \in 0 \dots \text{MaxActivity} \\
& \wedge \text{idleObserved} \in \text{BOOLEAN} \\
& \wedge \text{idleHandled} \in \text{BOOLEAN} \\
& \wedge \text{heartbeatPending} \in \text{BOOLEAN} \\
& \wedge \text{inputUsed} \in \text{BOOLEAN} \\
& \wedge \text{activityAdvancedAfterObservation} \in \text{BOOLEAN} \\
& \wedge \text{lastOutcome} \in \{ \text{"ready"}, \text{"heartbeat-accepted"}, \\
& \quad \text{"heartbeat-rejected"}, \text{"revoked"}, \text{"fenced"} \}
\end{aligned}$$

$$\text{RevocationIsMonotonic} \triangleq \text{everRevoked} \Rightarrow \neg \text{live}$$

$$\text{NewerActivityFencesOlderIdle} \triangleq \text{activityAdvancedAfterObservation} \Rightarrow \text{live}$$